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A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: If space is a vacuum, how do things like gravity and light move through it?

– Govar Rizgar, 12

Ah, two questions rolled into one! We love that at YSJOI. Well, first off, although space isn't actually a perfect vacuum, let us pretend it is for the purposes of this question. Light seems magical – massless photons traversing nothing? Well, it travels across nothing by *making* something to travel across – namely, a magnetic wave. You see, unlike sound – which relies on vibrating particles to be transferred – light is an electromagnetic wave, a family of transverse waves that don't need a medium to travel through. How? Light is a ripple made of oscillating electric and magnetic fields. These fields can sustain each other and move forward without needing a physical medium. When the Sun emits light, for example, it sends out these electromagnetic waves at the incredible and aptly named speed of light – $3x10^8$ m/s. They cross 150 million kilometers of space and then strike your eyes, allowing you to see daylight. So even though space is empty of matter, it is not empty of fields.

Gravity, on the other hand, is not like light – it doesn't even have to travel through space at all. According to Einstein's theory of general relativity, gravity is not a force being sent across space, nor is it matter being sent through it; instead, massive objects like planets and stars bend spacetime itself and cause the phenomenon we know as gravity. Spacetime is the fabric that combines space and time into one structure. When Earth orbits the Sun, it isn't being pulled by an invisible rope – it's following a curved path in warped spacetime, like a ball rolling into a depression while also constantly trying to escape it. You can actually think of spacetime like a stretched rubber sheet. Place a heavy ball on it, and the sheet curves. Roll a smaller ball nearby, and it will spiral inward – not because the big ball "pulled" it, but because the surface itself is curved. This curvature is not actually 'material' as we think of it, but a property of space itself instead. This means that the curvature exists even in a vacuum – in this case, space. Of course, this means, in a rather fun technicality – space isn't empty at all!

Question II: What is time made of?

– Zheena Fikri, 16

Time – magical, invisible, all powerful, deserving of several books in its own right. It feels like an invisible river, always flowing straight ahead – you can see its effects in a growing plant, a moving clock, or in the mirror every year as you grow older. But what is time *made* of? Is it a substance, like water or air? Is it a force, like gravity or friction?

Actually, it's neither of the two! Physicists see time as a dimension – specifically the mystical 4th dimension. Just as the first dimension gives us the X axis of a grid, the second gives us the Y axis and the third gives us the Z axis, the fourth dimension – time – gives us another coordinate – that is, when something occurred. Imagine describing to your friend a place where you want to meet. You would tell them the longitude, the latitude, the floor of the building you would meet on – and *when* you would meet them. You can think of the universe as the four-dimensional fabric of spacetime, where the three dimensions of space (length, width, height) are stitched together with time. In this sense, time isn't made of "stuff" – it is a measure of change, a way to order events from past to present to future.

Einstein showed us that time is flexible. It can stretch or shrink depending on how fast you move or how strong gravity is where you are. Clocks on satellites tick slightly faster than clocks on Earth, because the pull of Earth's gravity slows time down here on Earth in the movie-famous phenomenon of time-dilation. So time is like a shape-shifter, flowing differently for you depending on your perspective. Quantum physics complicates things even further, because some scientists suggest that at the tiniest scales – the scale of atoms and electrons and quarks – time might not be continuous at all. Instead, it could be made of tiny "time quanta," like little, singular frames of the movie we call reality. We can't yet perceive these tiny units, but they might govern the rhythm of the universe itself.

Time is not a force nor a thing; it doesn't measure atoms nor is it measured by them. Time is, in its very essence, change itself.

Question III: If evolution is random, how did different species evolve so many similar organs?

– Rwar Rebin, 14

Evolution often gets called "random," but that can be misleading. Random mutations happen in DNA, yes – small changes, mistakes, or quirks in genes, and these are truly random – copying errors during mitosis, for example. But evolution itself is not purely random; it's a mixture of chance and selection.

You can think of mutations like rolling dice. Most rolls don't help but are still harmless; some hurt and may leave you at a disadvantage – white fur in a tropical environment, for instance – but a lucky roll might create a tiny advantage – like slightly sharper claws, better eyesight, or a faster top speed. Nature then "selects" the winners: those individuals survive, reproduce, and pass on their genes, and because of their slight advantage over the others of the species, they eventually begin outnumbering the others. Over millions of years, these small advantages build into big changes, creating organs, limbs, and even entire body plans. So how do similar organs appear in extremely different species, like wings in bats and birds, or eyes in octopuses and humans? This is because of two things: convergent evolution and shared ancestry. Convergent evolution is when unrelated species face similar challenges and pressures – wings, for example, solve the problem of flight, helping prey flee extremely fast from predators and vice versa. Even though bats and birds are very different animals, evolution "found" the same solution independently. In some cases, however, the similarity comes from a shared ancestor, whose genes carried the blueprint for certain structures. Over time, descendants tweak these blueprints in different ways, but the basic design and largest organs stay similar.

Evolution is a crazy tinkerer who shapes life over millions of years into many forms, but often returns to the same simple solutions.

Question IV: If light has no mass how does gravity affect it?

– Yousif Abdullah, 12

Light and gravity – two things that feel like they simply can't mix. After all, it seems impossible, doesn't it? From what we know and were taught as kids: gravity pulls on things with mass and affects them, and light has no mass and is pure energy. So obviously, light should just ignore gravity and fly straight forever, right?

The answer to this actually ties back to what we learned back in question I of this issue: in Einstein's mind-blowing theory of general relativity, gravity isn't just a force that tugs on – it's the curvature of spacetime itself. Massive objects like stars and planets bend the fabric of space and time around them, and light always travels along the "straightest path" it can find. But in a curved space, the straightest path is…well, curved!

Even though light has no mass, it does travel, as it is a progressive electromagnetic wave! And when it moves through any space it must, of course, follow a path. It always moves in a straight line in spacetime, but spacetime itself is bent near massive objects, so the path of the light bends. This is called gravitational lensing – we can actually see it with telescopes as stars or galaxies appear shifted or even duplicated because their light bends around another massive object – which can be a real pain for scientists trying to discern information about objects!

And when objects are extremely dense, immense, and have so much gravity that they are a terrifying well of gravity – like black holes – they can even absorb the light and it's energy, giving the black hole its name and most famously scary characteristic!

So even without mass, light is affected by gravity because it follows the curves in spacetime. Gravity doesn't tug on the photons like a rope; instead, it changes the very path on which they travel, sometimes drawing them in so well that they simply can't escape.

And this shows us one of the universe's strangest truths: massless things are still bound to the geometry of space and time. No-one, not even those travelling at the literal speed of light, are exempt from the laws of the universe!

Question V: Could humans evolve to breathe on Mars?

– Arwan Ahmed, 15

Humans on Mars – already sounds like sci-fi, right? Already a milestone that we're struggling to reach. But could our species actually evolve to survive there, especially breathing the thin, mostly carbon dioxide atmosphere? Unfortunately for the mars enthusiasts – like us at YSJOI – the short answer is not naturally and certainly not quickly.

Evolution is tiringly slow. On Earth, humans evolved over millions of years in oxygen-rich air at sea level, and never really had a need to convert CO_2 into energy. Up until recently, actually, CO_2 was barely of note as a gas! Our lungs, blood, and even our DNA are tuned to that environment. Mars, on the other hand, has an atmosphere that's 100 times thinner than Earth's and is 95% carbon dioxide. Because of our years of fine-tuning for oxygen, that's deadly for humans. Breathing there without a spacesuit would be instant death. Because the pressure would simply not build – anyone who took off their suits would immediately die – them reproducing and producing children who might slowly adapt to the environment is out of the question.

Even today, at great, high peaks, only the rare individuals with genetic mutations that help tolerate low oxygen or higher CO_2 survive and reproduce. Over many generations, you might get humans with stronger lungs, hemoglobin that binds oxygen better, or even biochemical tweaks to handle carbon dioxide. On Mars, most of the things humans need to survive simply don't exist; before reproducing to allow evolution to occur, the human would have to eat, grow and develop. People just wouldn't make it long enough for natural selection to do its job.

That's why real plans for Mars rely on technology, not biology. Spacesuits, pressurized habitats, oxygen generators, and possibly engineered microbes to help humans breathe – these are our shortcuts to survival. Maybe in the far future, with genetic engineering, we could speed things up.

But for now, humans evolving to breathe on Mars naturally is about as close to impossible as it gets.

Acknowledgements

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For these incredible questions and giving us a chance to answer them in the December 2025 issue of YSJOI!

And we thank you, reader, for supporting our mission to help the Iraqi youth where there has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at www.YSJOI.is-great.org

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